

SETS, RELATIONS AND FUNCTIONS

Set:

A set is a collection of well-defined objects i.e. the objects follow a given rule or rules.

Elements of a set:

The members of a set are called its elements. If an element x is in set A , we say that x belongs to A and write $x \in A$. If the element x is not in A then we write $x \notin A$.

Examples of sets:

1. The set of vowels in the alphabet of English language.
2. The set of all points on a particular line.

Some special sets:

(i) Finite and infinite sets:

A set A is finite if it contains only a finite number of elements; we can find the exact number of elements in the set. Otherwise, the set is said to be an infinite set.

Example:

$$Q = \text{set of all rational numbers} = \left\{ \frac{p}{q} : p, q \in \mathbb{Z}, q \neq 0 \right\}$$

$$R = \text{set of all real numbers} = \{x : x \text{ is a rational and an irrational number}\}$$

$$C = \text{set of all complex numbers} = \{x + iy; \quad x, y \in \mathbb{R}\}$$

(ii) Null set:

A set which does not contain any element is called a null set and is denoted by ϕ . A null set is also called an empty set.

(iii) Singleton set:

A set which contains only one element is called a singleton set.

(viii) Power set:

The power set of a set A is the set of all of its subsets, and is denoted by $P(A)$ e.g. if $A = \{4, 5, 6\}$ then

$$P(A) = \{\phi, \{4\}, \{5\}, \{6\}, \{4, 5\}, \{5, 6\}, \{4, 5, 6\}\}.$$

Note: The null set ϕ and set A are always elements of $P(A)$.

Theorem: If a finite set has n elements, then the power set of A has 2^n elements.

Operations on sets:

The operations on sets, by which sets can be combined to produce new sets.

(i) Union of sets:

The union of two set A and B is defined as the set of all elements which are either in A or in B or in both. The union of two sets is written as $A \cup B$;

(ii) Intersection of sets:

(i) The intersection of two sets A and B is defined as the set of those elements which are in both A and B and is written as

$$A \cap B = \{x : x \in A \text{ and } x \in B\}$$

(ii) The intersection of n sets $A_1, A_2, \dots, A_n$ is written as

$$\bigcap_{i=1}^n A_i = A_1 \cap A_2 \cap A_3 \dots \dots \dots \cap A_n = \{x : x \in A_i \text{ for all } i, 1 \leq i \leq n\}.$$

Disjoint sets:

Two set A and B are said to be disjoint, if there is no element which is in both A and B , i.e. $A \cap B = \varphi$;

✓ The properties of the complement of sets are known as **DeMorgan laws**, which are

(i) $A^c - B^c = B - A$

(ii) $(A \cup B)^c = A^c \cap B^c$

(iii) $(A \cap B)^c = A^c \cup B^c$

✓ If A and B are not disjoint, then

(i) $n(A \cup B) = n(A) + n(B) - n(A \cap B)$

(ii) $n(A \cup B) = n(A - B) + n(B - A) + n(A \cap B)$

(iii) $n(A) = n(A - B) + n(A \cap B)$

(iv) $n(B) = n(B - A) + n(A \cap B)$

(vi) Cartesian product of sets:

Let a be an arbitrary element of a given set A i.e. $a \in A$ and b be an arbitrary element of B i.e. $b \in B$. Then the pair (a, b) is an ordered pair. Obviously $(a, b) \neq (b, a)$. The cartesian product of two sets A and B is defined as the set of ordered pairs (a, b) . The cartesian product is denoted by $A \times B$

$$\Rightarrow A \times B = \{(a, b); a \in A, b \in B\}.$$

Relation:

Let A and B be two sets. A relation R from the set A to set B is a subset of the Cartesian product $A \times B$. Further, if $(x, y) \in R$, then we say that x is R -related to y and write this relation as $x R y$. Hence $R = \{(x, y); x \in A, y \in B, x R y\}$.

Domain and Range of a relation: Let R be a relation defined from a A set to a set B , i.e. $R \subseteq A \times B$. Then the set of all first elements of the ordered pairs in R is called the domain of R . The set of all second elements of the ordered pairs in R is called the range of R .

That is, $D = \text{domain of } R$ and $R^* = \text{range of } R$.

$D = \text{domain of } R = \{x : (x, y) \in R\} \text{ or } \{x : x \in A \text{ and } (x, y) \in R\},$

$R^* = \text{range of } R = \{y : (x, y) \in R\} \text{ or } \{y : y \in B \text{ and } (x, y) \in R\}.$

Clearly $D \subseteq A$ and $R^* \subseteq B$.

FUNCTIONS:

A mapping $f: X \rightarrow Y$ is said to be a function if each element in the set X has its image in set Y . Every element in set X should have one and only one image.

Let $f: R \rightarrow R$ where $y = x^3$. Here for each $x \in R$ we would have a unique value of y in the set R .

Set ' X ' is called domain of the function ' f '.

Set ' Y ' is called the co-domain of the function ' f '.

Algebra of Functions:

Let us consider two functions,

$f: D_1 \rightarrow R$ and $g: D_2 \rightarrow R$. We describe functions $f + g$, $f - g$, $f.g$ and f/g as follows:

- $f + g : D \rightarrow R$ is a function defined by
- $(f + g)x = f(x) + g(x)$ where $D = D_1 \cap D_2$
- $f - g : D \rightarrow R$ is a function defined by
- $(f - g)x = f(x) - g(x)$ where $D = D_1 \cap D_2$
- $f.g : D \rightarrow R$ is a function defined by
- by $(f.g)x = f(x). g(x)$ where $D = D_1 \cap D_2$
- $f/g : D \rightarrow R$ is a function defined by
- $(f/g)x = \frac{f(x)}{g(x)}$ where $D = \{x : x \in D_1 \cap D_2, g(x) \neq 0\}$

TYPE OF FUNCTION

One-One and Many-One Functions:

When every element of domain of a function has a distinct image in the co-domain, the function is said to be One-One. If there are at least two elements of the domain whose images are the same, the function is known as Many-One.

Onto and Into Functions:

For every point y in b , there is some point x in A such the

$f(x) = y$. It is called onto function. When the codomain y which is not an image of any element in the domain x , then function is **onto**.

Even and Odd Functions:

- $f(x) - f(-x) = 0$ for even function and $f(x) + f(-x) = 0$ for odd functions .

Periodic Function

If $f(x)$ is periodic with period t , then $a f(x) + b$ where $a, b \in \mathbb{R}$ ($a \neq 0$) is also periodic with period t

Some Important function and their domain and range

FUNCTION F(X)	DOMAIN	RANGE
$\sin x$	$(-\infty, \infty)$	$[-1, 1]$
$\cos x$	$(-\infty, \infty)$	$[-1, 1]$
a^x $a > 1$	$\mathbb{R}$	$(0, \infty)$
$\log_a x$, $a > 0$ and $\neq 1$	$(0, \infty)$	$(-\infty, \infty)$
$[x]$	$\mathbb{R}$	$\mathbb{I}$
$ x $	$\mathbb{R}$	$[0, \infty)$
$\{x\}$	$\mathbb{R}$	$[0, 1)$